

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: HUT310

Course Name: MANAGEMENT FOR ENGINEERS

Max. Marks: 100

Duration: 3 Hours

PART A

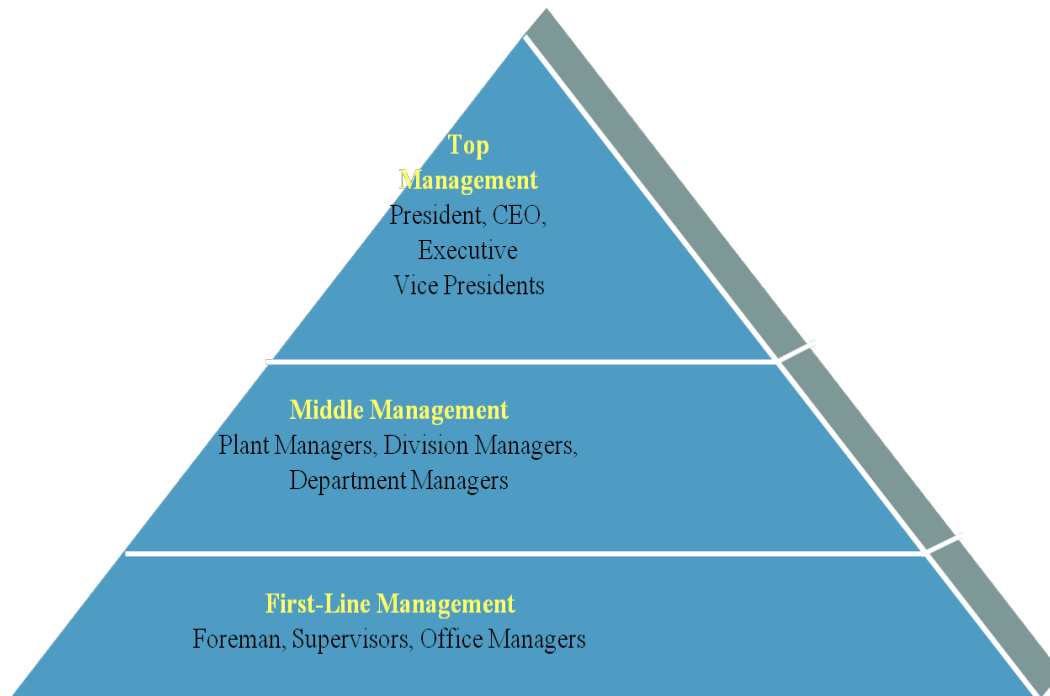
(Answer all questions; each question carries 3 marks)

Marks

1 What are the different levels of management?

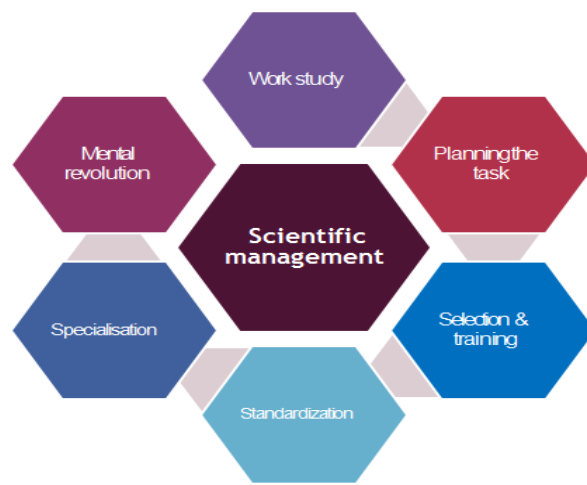
(3)

Ans: Bottom level, Middle Level, Top level



2 What is scientific management?

- ❖ According to Taylor, scientific management is an art of knowing exactly what you want your men to do and seeing that they do it in the best and cheapest way
- ❖ If a work is analysed scientifically it will be possible to find one best way out of it.
- ❖ In his book, Principles of scientific management, he formulated a view on management which was highly inspired by engineering principles.



3 Explain the importance of delegation in management.

Delegation of authority is a process in which the authority and powers are divided and shared amongst the subordinates. When the work of a manager gets beyond his capacity, there should be some system of sharing the work. This is how delegation of authority becomes an important tool in organization function. Through delegation, a manager, in fact, is multiplying himself by dividing/multiplying his work with the subordinates. The importance of delegation can be justified by -

1. Through delegation, a manager is able to divide the work and allocate it to the subordinates. This helps in reducing his work load so that he can work on important areas such as - planning, business analysis etc.
2. With the reduction of load on superior, he can concentrate his energy on important and critical issues of concern. This way he is able to bring effectiveness in his work as well in the work unit. This effectivity helps a manager to prove his ability and skills in the best manner.
3. Delegation of authority is the ground on which the superior-subordinate relationship stands. An organization functions as the authority flows from top level to bottom. This in fact shows that through delegation, the superior-subordinate relationship become meaningful. The flow of authority is from top to bottom which is a way of achieving results.
4. Delegation of authority in a way gives enough room and space to the subordinates to flourish their abilities and skill. Through delegating powers, the subordinates get a feeling of importance. They get motivated to work and this motivation provides appropriate results to a concern. Job satisfaction is an important criterion to bring stability and soundness in the relationship between superior and subordinates. Delegation also helps in breaking the monotony of the subordinates so that they can be more creative and efficient. Delegation of authority is not only helpful to the subordinates but it also helps the managers to

develop their talents and skills. Since the manager get enough time through delegation to concentrate on important issues, their decision-making gets strong and in a way they can flourish the talents which are required in a manager. Through granting powers and getting the work done, helps the manager to attain communication skills, supervision and guidance, effective motivation and the leadership traits are flourished. Therefore it is only through delegation, a manager can be tested on his traits.

5. Delegation of authority is help to both superior and subordinates. This, in a way, gives stability to a concern's working. With effective results, a concern can think of creating more departments and divisions flow working. This will require creation of more managers which can be fulfilled by shifting the experienced, skilled managers to these positions. This helps in both virtual as well as horizontal growth which is very important for a concern's stability.

- 4 "Planning is looking ahead and control is looking back" Comment (3)

In planning process the managers foresee the future ,the targets to be achieved and goal to be accomplished but in the case of controlling we do a checking process on the processes going on and do the correction.Feedback,feedforwards and concurrent control are the various schemes of controlling process.

- 5 How can we improve the productivity of an organisation? (3)

Productivity =Output/Input

We can improve productivity by either increasing the output or lowering the input and maintaining the same output.Improving productivity means either reducing the amount of materials and labor you're putting into the process, or increasing the level of output for the same quantity of input. That's productivity in a nutshell.

- 6 List out the steps in rational decision-making.

1.**Rational model**: The rational decision-making model suggests that you follow a sequence of steps in order to find a solution to the problem.

- start by defining the problem
- identify the criteria that you'll use to brainstorm possible solutions.
- decide how important each criterion is and create a list of possible alternatives.
- evaluate those alternatives and then determine the best solution.

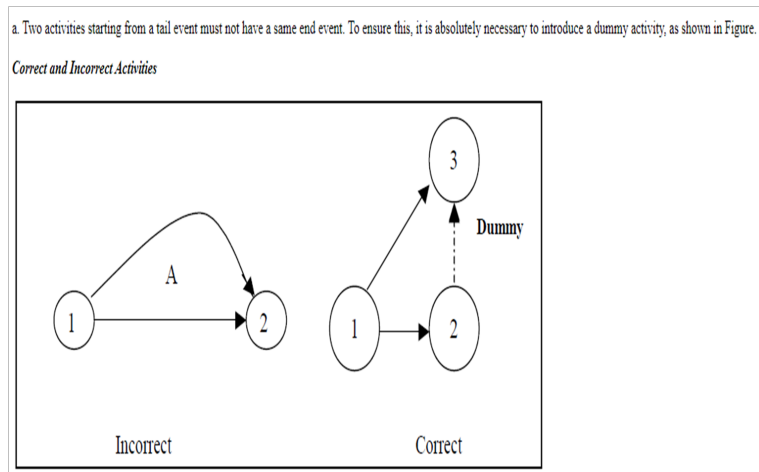
7

Differentiate between CPM and PERT.

Meaning	PERT is a project management technique, used to manage uncertain activities of a project.	CPM is a statistical technique of project management that manages well defined activities of a project.
What is it?	A technique of planning and control of time.	A method to control cost and time.
Orientation	Event-oriented	Activity-oriented
Evolution	Evolved as Research & Development project	Evolved as Construction project
Model	Probabilistic Model	Deterministic Model
Focuses on	Time	Time-cost trade-off
Estimates	Three time estimates	One time estimate

8

Explain the dummy activity in a project network.



9

Differentiate between needs, wants, and demands.

Needs: The basic requirements for human beings to survive.

Wants: Need when shaped by culture and individual personality becomes want.

Demands: When backed by buying power, want becomes a demand. (3)

Derived from French verb which means to undertake

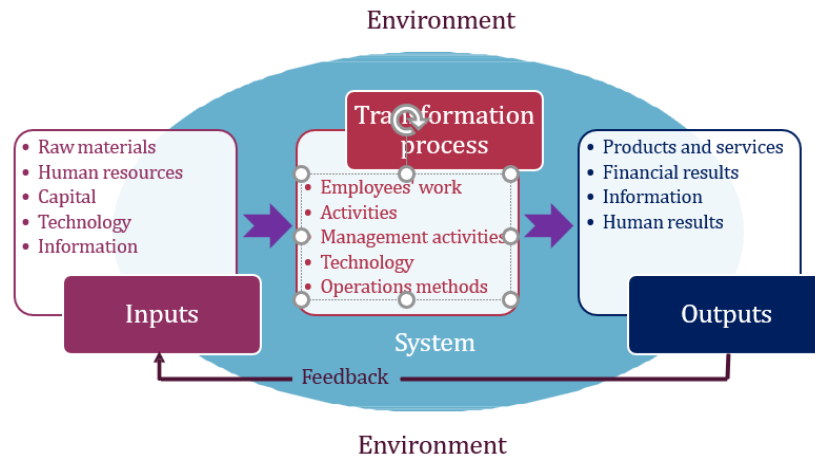
- This refers to those who undertake the risk of new enterprises- enterprise is created by an entrepreneur- and the process is called entrepreneurship
- Always in search of new ideas- creative , innovative
- Ready to accept the risk and uncertainty with the enterprise

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11 a) Explain the systems approach to management with a block diagram. (10)
- ❖ The systems approach of management states that organizations represent a complex collection of various components that work together to reach a common goal. An organization is made up of numerous subsystems, such as different departments.
 - ❖ The systems approach implies **that decisions and actions in one organizational area will affect other areas**. For example, if the purchasing department does not acquire the right quantity and quality of inputs, the production department wont be able to do its job.
 - ❖ The approach is based on the concept that organizations can be visualized as systems.
 - ❖ A system is a set of interrelated parts that operate as a whole to achieve common goals.
 - ❖ The full system is made up of many parts. Each of these parts is called a subsystem. A system may be a subsystem of a larger system.
 - ❖ A system can either be open or closed.
 - ❖ Output of a system is always more than the combined output of its parts. This is called synergy.
 - ❖ In organizational terms, synergy means that that as separate departments within organization cooperate and interact, they become more productive than if they had acted in isolation
 - ❖ An organization is a system consisting of many interrelated and interdependent sub-systems.
 - ❖ The sub-parts should be studied in their inter-relationships rather than in isolation.
 - ❖ The organization is responsive to environmental effect.



• **5 main Advantages of Systems Approach in management**

- It concentrates on end results rather than the means.
- It provides an orderly and efficient plan of action. ...
- It develops coordination of the specialized activities.
- It provides a good basis of control.
- It frees management from many daily details of operations management.

b) How are changes in the external environment compensated for in the systems approach?

(4)

External environment factors are important because they **can cause direct and indirect effects on business operations, personnel and revenue**. The external environment of a company changes constantly in ways beyond the company's control, but executives and managers can track these changes and minimize their consequences

As a system, an organization draws inputs (energy, info, materials etc.) from its environment. It transforms these inputs and returns the output into the environment in the form of goods and services. Organization is an open system and it interacts with its environment

12 a) Explain Task and Responsibilities of a professional Manager.

(8)

Envisioning Goals
 Managing Growth
 Improving and Maintaining Efficiency
 Innovation
 Looking out for the competition
 Leadership
 Change Management
 Choosing correct Information Technology
 To create a surplus

Managers must raise productivity

- b) Explain the importance of management in an organisation. (6)

Benefits of specialisation: An organisation avails the benefits of division of labour and specialisation by allocating jobs to the individuals who are specialised in performing that jobs.

Clarity in working relationships: By establishing working relationships, an organisation clarifies the responsibilities and authorities of individuals in different departments . This minimises the conflict and confusion about the respective powers and privileges of managers in an organisations.

Optimum utilisation of resources: Organising leads to the proper utilisation of resources- financial and human resources. An organisation can avoid overlapping of work and can make the best use of resources by assigning jobs to the employees properly.

Adaptation to Change: The process of organising allows an organisation to accommodate changes in the business environment

Effective Administration: Organising helps to avoid confusion and duplication in a business enterprise by providing clear description of jobs and related duties.

6. Development of personnel : Due to the effective delegation of authority and proper division of work, the employees are assigned with the tasks according to their specialisation. Employees in this way develop new methods for developing their tasks which increases their efficiency and leads to their development.

7. Expansion and Growth: Helps in growth and expansion of the organisation

Module -2

- 13 a) Explain the mission, goals, strategy, organisation with suitable examples. (8)

- ❖ **Goals of an organization** are derived from the mission of an organization. Goals are more specific than mission but address, in general, key issues as market share, productivity, profitability, efficiency, growth, physical and financial resources etc. Goals are the broad and long term targets of an organization
- ❖ **Mission or purpose** identifies the basic purpose or function or tasks of an enterprise. its the official goal that describe an organization's reason for existence. Time span and specificity are the two factors that distinguish objectives from goals.
- ❖ **Strategy** is an outline of steps designed with the goals of the entire organization as a whole, rather than with the goals of specific departments. It is a plan that drives organization's efforts towards goal accomplishment. It look ahead over the next two or even more years to move the organization from where it currently is to where it wants to be

Organisation is a mechanism or structure which helps the activities to be performed effectively.

The primary purpose of organizing is to bring order to the organization so that there is no confusion in the conduct of the business

b) Differentiate between strategic and tactical decisions.

(6)

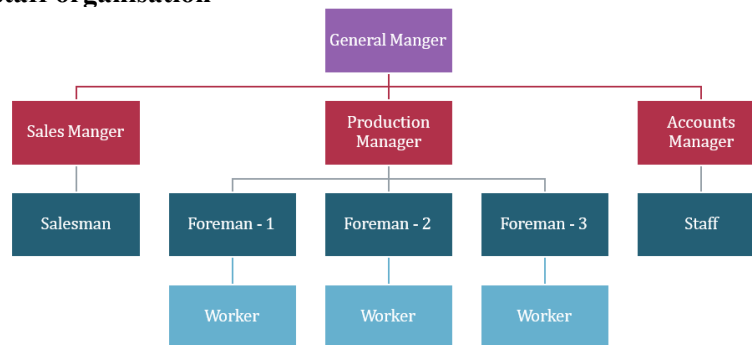
Strategic Plan is an outline of steps designed with the goals of the entire organization as a whole, rather than with the goals of specific departments. It is a plan that drives organization's efforts towards goal accomplishment. It look ahead over the next two or even more years to move the organization from where it currently is to where it wants to be. Top level management develops the strategic plan for the entire organization, while lower levels of management develops compatible objectives and plans to achieve them.

Tactical Plan is the plan designed to implement the strategic goals and plans formulate the top management. It is the responsibility of the middle managers to take the broad strategic plan and identify the specific tactical actions. Tactical plans are the organized sequence of steps designed to execute strategic plans. The scope of tactical plans is generally broader than operational plans but is narrower than that of strategic plans. These plan generally span one year or less. Focus on functional areas of an organization like marketing, manufacturing, financing etc.

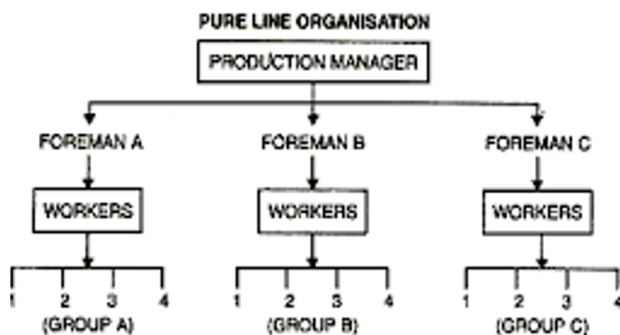
14 a) Illustrate the different types of organisation structures.

(8)

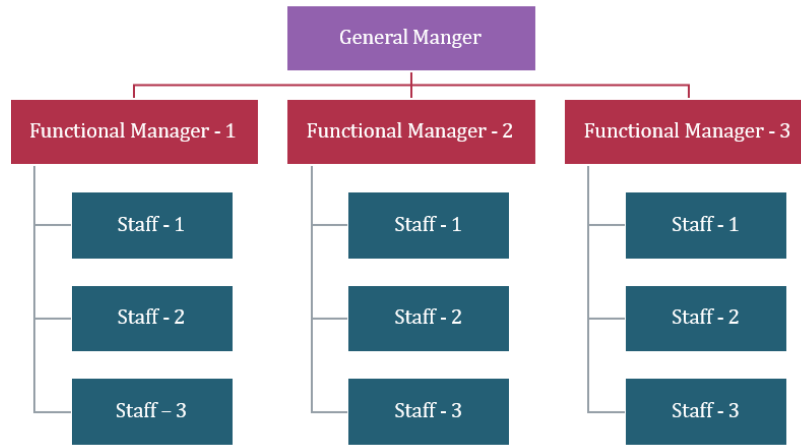
Line & staff organisation



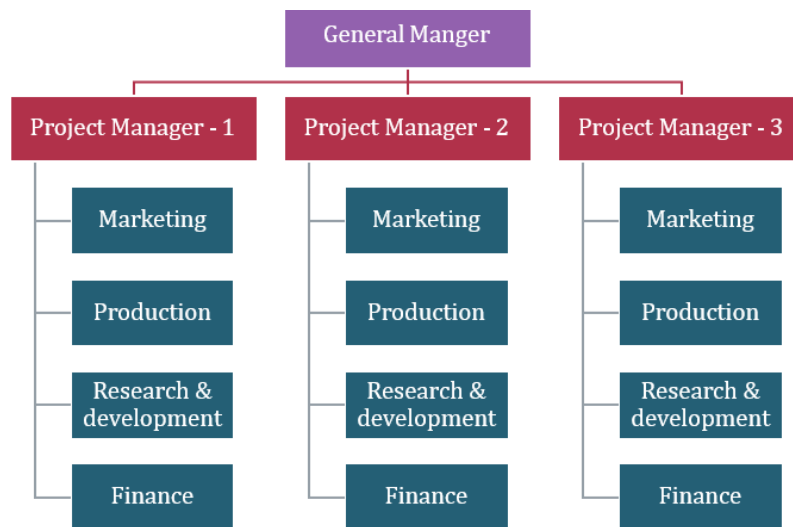
Line organisation



Functional organisation



Project Organasation



b) Explain the factors governing the selection of organisation structures.

(6)

- 1.Unity of objectives
- 2.Clear line of authority and responsibility
- 3.Division of work and specialization
- 4.Span of control
- 5.Unity of command
- 6.Simple and flexible

- 15 a) The producer of an apple crates company produces 270 crates per 100 logs with his current equipment. He currently purchases 100 logs per day, and each log required 3 labour hours to process. He believes that he can hire a professional buyer who can buy a better quality log at the same cost. If this is the case, he increases his production to 290 crates per 100 logs. His labour hours will increase by 8 hours per day. What will be the impact on productivity (measured in crates per labour-hour) if the buyer is hired? What is the growth in productivity in this case?

15a) Labour productivity for the current method:-

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}} = \frac{270 \text{ crates}}{100 \text{ logs} \times 3 \text{ hrs./log}} = 0.9 \text{ crates/labour-hr}$$

The current labour hours is 300 labour hours/day. Buying better quality logs will increase the output to 290 crates per 100 logs & increase the labour hours to 308 labour hours/day. Labour productivity in this case is calculated as follows.

$$\text{Productivity} = \frac{290 \text{ crates}}{(100 \text{ logs} \times 3 \text{ hrs./log}) + 8 \text{ hrs}} = 0.942 \text{ crates/labour-hr}$$

This means that the labour productivity would increase by $\left(\frac{0.942 - 0.9}{0.9} \right) \times 100 = 4.67\%$

- b) Explain decision making under uncertainty.

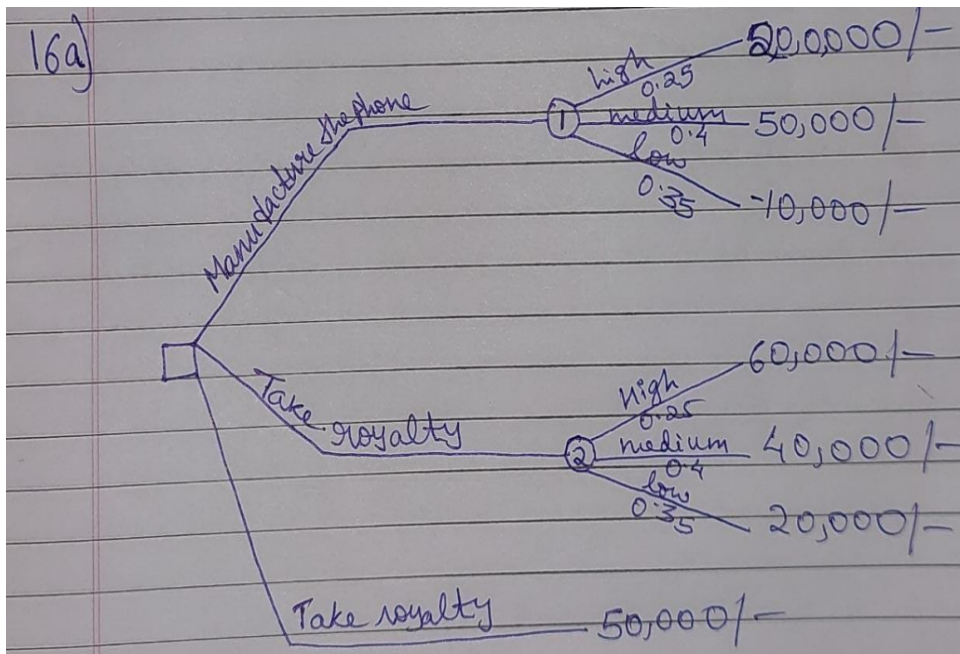
(4)

- ❖ At times a decision maker cannot assess the probability of occurrence for the various states of nature. Uncertainty occurs when there exist several future states of nature but the probabilities of each of these states occurring are not known.
- ❖ Different approaches to decision making under uncertainty are:
 1. Maxi-max solution
 2. Maxi-min solution
 3. Mini-max solution
 4. Laplace criterion

- 16 a) A cell phone manufacturer has invented a 3D phone. The company wants to take decision whether to manufacture the phone, take royalty from another manufacturer, or sell rights of the invention and take a lump sum amount of ₹50,000. The profits associated and probability of these alternatives is given in the table below. Represent the problem as a decision tree and suggest a decision to

maximise profits.

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Demand	Manufacture the phone		Take royalty	
	Probability	Profit (₹)	Probability	Profit (₹)
High	0.25	200,000	0.25	60,000
Medium	0.4	50,000	0.4	40,000
Low	0.35	-10,000	0.35	20,000

$$(EMV)_{\text{Manufacture the phone}} = (2,00,000 \times 0.25) + (50,000 \times 0.4) + (-10,000 \times 0.35)$$

$$= 66,500/-$$

$$(EMV)_{\text{Take royalty}} = (60,000 \times 0.25) + (40,000 \times 0.4) + (20,000 \times 0.35)$$

$$= 38,000/-$$

$$(EMV)_{\text{Sell rights of invention}} = 50,000/-$$

Hence the optimal decision for the company to maximise profits is to manufacture the phone.

- b) Explain the different models of decision-making behaviour.

The rational model counteracts a lot of the factors – like faulty assumptions – that can lead us to bad decisions. It can minimize risk and uncertainty. This model is also one you can use on your own or as part of a team.

However, it's not the best model to use when you're under time constraints or in a fast-changing situation. It's also important to remember that you won't always have all the information you need to use this model. And, even if you do, going through the full process isn't efficient or necessary for some decisions.

Bounded rationality decision-making model bounded rationality when you don't have enough time or information to follow the full rational decision-making model. Sometimes it's better to have a good enough decision sooner vs. a "perfect" decision that's delayed. And it burns a lot less mental energy and other resources.

Vroom-Yetton Decision-Making Model There's no one ideal process for making decisions. Instead, the best process to use will change based on your situation. That's the idea behind the Vroom-Yetton decision model (sometimes known as Vroom-Yetton-Jago). The

first part of this model uses seven yes-or-no questions. Your answers to the questions then guide you toward one of five decision-making processes to use. Options range from making the decision based on what you know now without consulting your team to reaching a group consensus with your team.

Module -4

17a) project consists of 7 activities and the three time estimates are as follows:

Activities	Duration in weeks		
	t_o	t_m	t_p
A	2	6	10
B	4	6	12
C	2	3	4
D	2	4	6
E	3	6	9
F	6	10	14
G	1	3	5

The sequence of activities is as follows:

- Activities *A* and *B* start at the beginning of the project.
 - When *A* is completed *C* and *D* start.
 - *F* can start when *B*, *C* and *D* are completed and is the final activity.
 - *G* can start when *F* is finished and is the final activity.
- i. What is the expected time of the duration of the project?
 - ii. Identify the critical path of the project.
 - iii. Calculate the probability that project will be completed in 23 weeks.

17a) Estimated time, $t_e = \frac{t_o + (4 \times t_m) + t_p}{6}$

Activities	Duration (in weeks) (t_e)
A	6
B	3
C	4
D	6
E	10
F	3

b) Write notes on Fulkerson's rule of numbering events. (4)

1. The initial event which all outgoing arrows with no incoming arrow is numbered 'a'
2. Delete all the arrows coming out from node '1'
3. Delete all the arrows going out from these numbered events to create more initial events
4. Continue until the final or terminal node which has all arrows coming in, with no arrow going out is numbered.

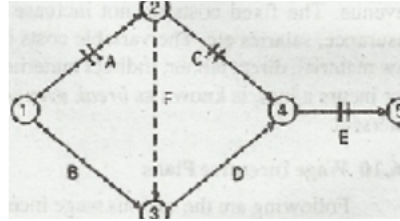
c) a) The following table shows the precedence requirements, normal and crash times, and normal and crash costs for a project. The indirect costs are ₹ 70/day. (10)

Activity	Required Time (days)		Cost (₹)	
	Normal	Crash	Normal	Crash
1-2	8	6	100	200
1-3	4	2	150	350
2-4	2	1	50	90
2-5	10	5	100	400
3-4	5	1	100	200
4-5	3	1	80	100

- Draw the network.
- Determine the critical path.

- iii. Determine the optimal duration and the associated cost after crashing.
- b) Differentiate between activity and event in a project network. (4)

Event. The event is a specific instant of time which makes the start and the end of an activity. The event consumes neither time nor resources. It is represented by a circle and the event number is written within the circle, as shown in Fig.



Activity. Every project consists of a number of job operations or tasks which are called activities. An activity is shown by an arrow and it begins and ends with an event. An activity consumes time and resources. An activity may be performed by an individual or a group of individuals.

The activity may be classified as *critical activity*, *non-critical activity* and *dummy activity*. The activity is called critical if its earliest start time (E S T) plus the time taken by it, is equal to the latest finishing time (L F T). A critical activity (e.g. A, C or E) is marked either by a thick arrow or by two lines as shown in Fig

Module -5

19a) What are intellectual property rights? Explain the business importance of patents.

Intellectual Property Rights (IPR) gives them this protection, as well as helping them exploit and control their IP.

“The exclusive right granted by State, to prevent others from using, manufacturing, distributing - inventions, processes, applications, new and original designs, trademarks, new plant varieties, data bases and artistic and literary works”. Such a person is known as ‘rights owner’ or ‘rights holder’

Industrial Designs: Design deals with features, shapes, patterns, etc., applied to an article by an industrial process, manual or mechanical.

Eg., chair is a utility item. However, chair itself does not qualify for IPR, but its special carvings, embossing etc., is done which increases the value of chair though it’s utility remains same, it becomes eligible for IPR under Designs Act. Designs can be registered based on its originality, henceforth they can use ® or registered, with registration number.

Patents: Is a monopoly right granted to a person, who invented a new product or process of making an article, for **20years** under the **Indian Patens Act, 1970**, and can be renewed after expiration of period.

The inventor has to file for patent first, and then make his/ her invention to public. A patent has to be applied in each country by the inventor, to claim his rights in that country

b) Explain the various interrelationships between the following functional areas.

(i) Production and Marketing

The link between marketing and production is a **dual one**. On one hand, production capabilities determine the number and type of products which can be marketed, and, on the other hand, a more accurate prediction of sales forecast for each product and product line is essential for efficient production operations

(ii) Production and Finance

Production department's main duty is to produce the goods. For producing goods, it needs raw material, labor and other expenses. For paying all expenses, production department needs money and fund which will be fulfilled by finance department.

(iii) Production and Personnel.

The skilled labour or efficient personnel will increase the quality of products as well as quantity of products

20 a) Explain the process of market segmentation. How is the marketing mix related to market segmentation?

- Market segmentation is a marketing concept which divides the complete market set up into smaller subsets comprising of consumers with a similar taste, demand and preference.
- A market segment is a small unit within a large market comprising of like minded individuals.
- One market segment is totally distinct from the other segment.
- A market segment comprises of individuals who think on the same lines and have similar interests.
- The individuals from the same segment respond in a similar way to the fluctuations in the market.
- Psychographic segmentation

The basis of such segmentation is the lifestyle of the individuals. The individual's attitude, interest, value help the marketers to classify them into small groups.

- Behaviouralistic Segmentation

The loyalties of the customers towards a particular brand help the marketers to classify them into smaller groups, each group comprising of individuals loyal towards a particular brand.

- Geographic Segmentation

Geographic segmentation refers to the classification of market into various geographical areas. A marketer can't have similar strategies for individuals living at different places.

The division of a market into different homogeneous groups of consumers is known as market segmentation. Rather than offer the same marketing mix to vastly different customers, market segmentation makes it possible for firms to tailor the marketing mix for specific target markets, thus better satisfying customer needs.



b) “Operations management is process of planning, organizing and controlling the activities of a production function”. Explain.

Operations management is an area of management concerned with designing and controlling the process of production and redesigning business operations in the production of goods or services. It involves the responsibility of ensuring that business operations are efficient in terms of using as few resources as needed and effective in meeting customer requirements.

It is concerned with managing an entire production or service system which is the process that converts inputs (in the forms of raw materials, labor, consumers, and energy) into outputs (in the form of goods and/or services for consumers). Operations produce products, manage quality and create services. Operation management covers sectors like banking systems, hospitals, companies, working with suppliers, customers, and using technology. Operations is one of the major functions in an organization along with supply chains, marketing, finance and human resources. The operations function requires management of both the strategic and day-to-day production of goods and services

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